

Mathematics

Classes Nine-Ten



NATIONAL CURRICULUM AND TEXTBOOK BOARD, BANGLADESH

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Mathematics

Classes Nine-Ten

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Preface

The aim of secondary education is to make the learners fit for entry into higher education by flourishing their latent talents and prospects with a view to building the nation with the spirit of the Language Movement and the Liberation War. To make the learners skilled and competent citizens of the country based on the economic, social, cultural and environmental settings is also an important issue of secondary education.

The textbooks of secondary level have been written and compiled according to the revised curriculum 2012 in accordance with the aims and objectives of National Education Policy-2010. Contents and presentations of the textbooks have been selected according to the moral and humanistic values of Bengali tradition and culture and the spirit of Liberation War 1971 ensuring equal dignity for all irrespective of caste and creed of different religions and sex.

The present government is committed to ensure the successful implementation of Vision 2021. Honorable Prime Minister, Government of the People's Republic of Bangladesh, Sheikh Hasina expressed her firm determination to make the country free from illiteracy and instructed the concerned authority to give free textbooks to every student of the country. National Curriculum and Textbook Board started to distribute textbooks free of cost since 2010 according to her instruction.

Mathematics plays an important role in developing scientific knowledge at this time of the 21st century. Not only that, the application of Mathematics has increased in family and social life including personal life. With all these things under consideration Mathematics has been presented easily and nicely at the Secondary level to make it useful and delightful to the learners, and a good number of new topics have been included in the textbook.

Considering the challenges and commitments of 21st century and following the revised curriculum the textbook has been written. The textbook has been revised and reedited by a group of prominent educationists to make it learner friendly in 2017. I thank sincerely all for their intellectual labor who were involved in the process of revision, writing, editing, art and design of the textbook.

Prof. Narayan Chandra Saha

Chairman

National Curriculum and Textbook Board, Bangladesh

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Chapter 1

Real Numbers

History of numbers is as old as that of human civilization. Mathematics started with expressing quantity using numbers. According to Greek Philosopher Aristotle formal beginning of mathematics started with its practice among ancient Egyptian priests. So it can be said that number based Mathematics was created some two thousand years before the birth of Jesus Christ. Numbers and operations on them have reached today's universal form after that through the hands of different nations and civilizations.

Indian mathematicians introduced for the first time the concept of 0 and decimal number system for satisfying the need of counting natural numbers. This is considered as a milestone in the representation of numbers. Later on Indian and Chinese mathematicians expanded the concepts of 0, negative, real, integer and fractional numbers which were taken as the basis by Arabian mathematicians of medieval age. Muslim mathematicians of the middle east are given the credit of expressing numbers using decimal fractions. Again they first introduced irrational numbers in the form of square roots for solution of quadratic algebraic equations in 11th century. Historians think that around 500 A.D. Greek philosophers felt necessity of irrational numbers, especially square root of 2, for the purpose of geometrical drawing. In the nineteenth century European mathematicians gave completeness to real numbers through introducing complete set of rules of operations. In order to satisfying daily needs students are required to have clear understanding about real numbers. In this chapter real numbers have been discussed in general. At the end of this chapter students will learn —

- ▶ How to classify real numbers.
- ▶ How to deduce approximate values of real numbers by expressing it in decimal fractions.
- ▶ How to classify decimal fraction.

- How to interpret repeated decimal numbers and convert fractions into repeated decimal numbers.
- How to convert repeated decimal fractions into fractions.
- How to interpret infinite nonrepeating fractional numbers.
- How to interpret similar and dissimilar decimal fractions.
- How to carry out addition, subtraction, multiplication and division operations on repeated decimal fractions, and will be able to solve different relevant problems.

Classification of Real Numbers

Natural Number: 1, 2, 3, 4, ... etc are natural numbers. 2, 3, 5, 7, ... are primes and 4, 6, 8, 9, ... etc are composite numbers. If GCD (Greatest Common Divisor) of two integers is 1 then they are called mutually primes. For example 6 and 35 are mutually primes since they do not have a common division other than 1.

Integer: All positive and negative fractionless numbers are said to be integers. For example, ..., -3, -2, -1, 0, 1, 2, 3, ... etc are integers.

Fractional Number: Numbers expressible as $\frac{p}{q}$ are said to be fractional numbers where $q \neq 0$ and $q \neq 1$. For example, $\frac{1}{2}$, $\frac{3}{2}$, $\frac{-5}{3}$, $\frac{4}{6}$ etc are simple fractional numbers. In case of a simple fraction $\frac{p}{q}$ if $p < q$ then the fraction is called proper and if $p > q$ then the fraction is termed improper. For example $\frac{1}{2}$, $\frac{1}{3}$, $\frac{2}{3}$, $\frac{1}{4}$, ... etc are proper fractions. $\frac{3}{2}$, $\frac{4}{3}$, $\frac{5}{3}$, $\frac{5}{4}$, ... are improper fractions.

Rational Number: Numbers of the form $\frac{p}{q}$ are called rational when p and q are integers and $q \neq 0$. For example, $\frac{3}{1} = 3$, $\frac{11}{2} = 5.5$, $\frac{5}{3} = 1.666$, ... etc are rational numbers. Any rational number can be expressed as ratio of two mutually prime numbers. All integers and fractions are rational numbers.

Irrational Number: If a number cannot be expressed in the form $\frac{p}{q}$ where p and q are integers and $q \neq 0$, then it is called an irrational number. Root of a natural number not equal to perfect square or its fraction is an irrational number.

For example, $\sqrt{2} = 1.414213\dots$, $\sqrt{3} = 1.732\dots$, $\frac{\sqrt{5}}{2} = 1.118\dots$, are irrational numbers. No irrational number can be expressed as ratio of two integers.

Decimal Fractional Number: Rational and irrational numbers when expressed using decimal sign is called decimal fractional number. For example, $3 = 3.0$, $\frac{5}{2} = 2.5$, $\frac{10}{3} = 3.3333\dots$, $\sqrt{3} = 1.732\dots$, etc are decimal fractional numbers. If there are finite number of digits after decimal sign then this is called finite decimal fraction and if there are infinitely many digits then it is called infinite decimal fractions. For example, 0.52 , 3.4152 etc are finite decimal fractions, whereas $\frac{4}{3} = 1.333\dots$, $\sqrt{5} = 2.123512367\dots$ etc are infinite decimal fractions. Again, in case of infinite decimal numbers if some digits after decimal point are repeated they are called infinite repeated decimal fractions. If digits are not repeated they are called infinite decimal numbers without repetition. For example, $\frac{122}{99} = 1.2323\dots$, $5.1\dot{6}5\dot{4}$ etc are infinite repeated decimal fractions and $0.523050056\dots$, $2.12340314\dots$ etc are infinite decimal numbers without repetition.

Real Number: All rational and irrational numbers are called real numbers. For example, the following numbers are real numbers.

$$0, \pm 1, \pm 2, \pm 3, \dots \qquad \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{4}{3}, \dots$$

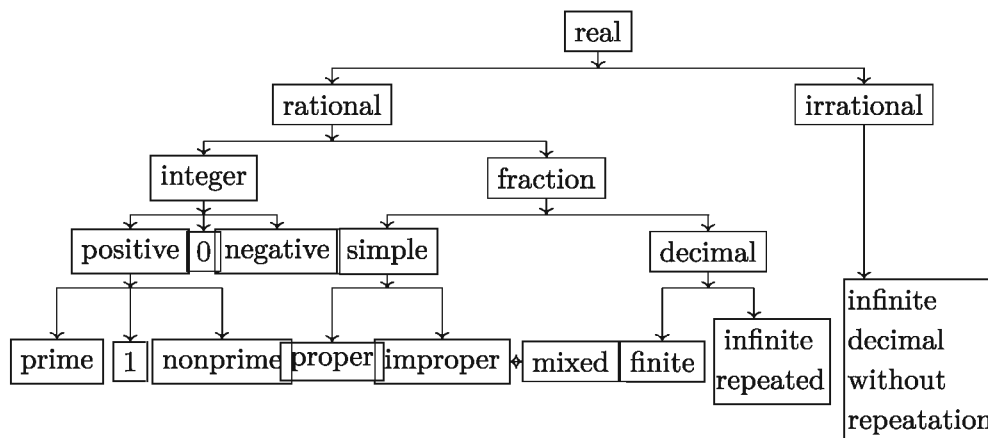
$$\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \dots \qquad 1.23, 0.415, 1.3333\dots, 0.\dot{6}\dot{2}, 4.120345061\dots$$

Positive Number: Numbers bigger than 0 are called positive numbers. For example, $2, \frac{1}{2}, \frac{3}{2}, \sqrt{2}, 0.415, 0.\dot{6}\dot{2}, 4.120345061\dots$, etc are positive numbers.

Negative Number: Numbers smaller than 0 are called negative numbers. For example, $-2, -\frac{1}{2}, -\frac{3}{2}, -\sqrt{2}, -0.415, -0.\dot{6}\dot{2}, -4.120345061\dots$, etc are negative numbers.

Non-negative Number: Positive numbers including 0 are called nonnegative numbers. For example, $0, 3, \frac{1}{2}, 0.612, 1.\dot{3}, 2.120345\dots$, etc are nonnegative numbers.

In the following chart we can see classification of real numbers.



Work: Show the position of the following numbers in the classification of real numbers $\frac{3}{4}$, 5, -7, $\sqrt{13}$, 0, 1, $\frac{9}{7}$, 12, $2\frac{4}{5}$, 1.1234, 0.323

Example 1. Find an irrational number between $\sqrt{3}$ and 4.

Solution: Here, $\sqrt{3} = 1.7320508 \dots$

Suppose, $a = \frac{\sqrt{3} + 4}{2} \approx 2.866$ and $b = \frac{\sqrt{3} + 4 + 4}{3} \approx 3.244$

Obviously a and b are both real numbers and both are bigger than $\sqrt{3}$ and smaller than 4 because a is average of two unequal numbers $\sqrt{3}$ and 4, and b is average of numbers $\sqrt{3}$, 4 and 4.

That is, $\sqrt{3} < 2.866 < 4$ and $\sqrt{3} < 3.244 < 4$

Again, a and b cannot be expressed as fractions.

$\therefore a$ and b are the desired numbers.

In fact we can construct countless such irrational numbers.

Basic characteristics of addition and multiplication processes of real numbers :

1. If a and b are real numbers (i) $a + b$ is a real number and (ii) ab a real number.
2. If a and b are real numbers (i) $a + b = b + a$ and (ii) $ab = ba$
3. If a , b , c are real numbers (i) $(a + b) + c = a + (b + c)$ and (ii) $(ab)c = a(bc)$
4. If a is real number, there are only two real numbers 0 and 1 such that (i) $0 \neq 1$, (ii) $a + 0 = 0 + a = a$ and (iii) $a \cdot 1 = 1 \cdot a = a$

5. If a is real number, (i) $a + (-a) = 0$ (ii) If $a \neq 0$ then $a \cdot \frac{1}{a} = 1$
6. If a, b, c are real numbers, $a(b + c) = ab + ac$
7. If a, b are real numbers $a < b$ or $a = b$ or $a > b$
8. If a, b, c are real numbers and $a < b$ then $a + c < b + c$
9. If a, b, c are real numbers and $a < b$ then (i) $ac < bc$ whenever $c > 0$
(ii) $ac > bc$ whenever $c < 0$

Proposition: $\sqrt{2}$ is an irrational number.

Proof: Suppose $\sqrt{2}$ is an irrational number.

Then there are two mutually prime numbers $p, q > 1$ so that $\sqrt{2} = \frac{p}{q}$

Or, $2 = \frac{p^2}{q^2}$ [squaring] That is $2q = \frac{p^2}{q}$ [Multiplying both sides by q]

Obviously $2q$ is an integer but $\frac{p^2}{q}$ is not an integer, because p and q are natural numbers and they are mutually prime and $q > 1$.

$\therefore 2q$ and $\frac{p^2}{q}$ cannot be equal. That is $2q \neq \frac{p^2}{q}$.

$\therefore \sqrt{2}$ cannot be expressed in the form $\frac{p}{q}$. That is $\sqrt{2} \neq \frac{p}{q}$

$\therefore \sqrt{2}$ is an irrational number. □

Work: Prove that $\sqrt{3}$ is an irrational number.

Example 2. Prove that when one is added to the product of 4 consecutive natural numbers this is a perfect square.

Solution: Let the consecutive integers be $x, x + 1, x + 2, x + 3$

If we add 1 to the product of these 4 integers we get:

$$\begin{aligned}
 & x(x+1)(x+2)(x+3) + 1 \\
 &= x(x+3)(x+1)(x+2) + 1 \\
 &= (x^2 + 3x)(x^2 + 3x + 2) + 1 \\
 &= a(a+2) + 1 \text{ [Assume } x^2 + 3x = a \text{]} \\
 &= a^2 + 2a + 1 = (a+1)^2 = (x^2 + 3x + 1)^2
 \end{aligned}$$

which is a perfect square. Therefore when 1 is added to the product of any four consecutive natural numbers this is a perfect square.

Decimal Fractions

Each real number can be expressed in decimal fractions. For example, $2 = 2.0$, $\frac{2}{5} = 0.4$, $\frac{1}{3} = 0.333\dots$ etc. Decimal fractions are of three types, namely: finite, repeated and infinite decimal fractions.

Finite decimal fractions: In any finite decimal fraction there is only a finite numbers of digits after the decimal point. For example, 0.12, 1.023, 7.832, 54.67, ... etc are finite decimal fractions.

Repeated decimal fractions: In any repeating decimal fraction all or some digits on the right of decimal point come again and again. For example, $3.333\dots$, $2.454545\dots$, $5.12765765\dots$ etc are repeated decimal fractions.

Infinite decimal fractions: In an infinite decimal fraction digits on the right of decimal point or its part will not appear repeatedly. For example, $1.4142135\dots$, $2.8284271\dots$ etc are Infinite decimal fractions.

Remarks: Finite and repeating decimal fractions are rational numbers whereas infinite decimal numbers are irrational numbers. Value of any irrational number can be computed to any desired decimal place. If numerator and denominator of a fraction can be represented by natural numbers then the fraction is a rational number.

Work: Classify the following numbers with reasons
 1.723 , $5.2333\dots$, 0.0025 , $2.1356124\dots$, $0.01050105\dots$ and $0.450123\dots$

Repeating decimal fractions

6) $23(3.833$

$$\begin{array}{r}
 18 \\
 \hline
 50 \\
 48 \\
 \hline
 20 \\
 18 \\
 \hline
 20 \\
 18 \\
 \hline
 2
 \end{array}$$

Express the fraction $\frac{23}{6}$ into decimal fraction. Observe that when numerator was divided by denominator the division process could not be completed. Observe that in the result the digit 3 appears over and over again. So $3.8333\dots$ is a repeating decimal number.

The decimal fraction for which the same digit or a set of digits in order appear after decimal point is called repeating decimal fraction. The part of the repeating decimal fraction that appears over and over again is called repeating. The remaining part is nonrepeating.

In a repeating decimal fraction if a single digit repeats then a point is marked above that digit. Otherwise point is marked on the first and the last repeating digits. For example, $2.555\dots$ is written as $2.\dot{5}$ and $3.124124124\dots$ as $3.\dot{1}2\dot{4}$.

In a decimal number if every digit after decimal point is repeating then it is called a pure repeating fraction. Otherwise the fraction is called mixed. For example, $1.\dot{3}$ is a pure repeating fraction and $4.2351\dot{1}\dot{2}$ is a mixed repeating fraction.

If in the denominator there are digits other than 2, 5 as a factor then no numerator will be divisible by that denominator. Since there cannot be any digit other than 1, 2, $\dots$, 9 at some stage as residue the same numbers will appear. Number of digits in the repeating part is always less than the number in the denominator.

Example 3. Express $\frac{3}{11}$ and $\frac{95}{37}$ into decimal fractions.

Solution: $\frac{3}{11}$ and $\frac{95}{37}$ have been converted into decimal fractions as below.

Actually 3 has been divided.
But since 3 is smaller than 11, 0 has been placed in the dividend and a 0 has been placed to the right of 3

$$11) 30(0.2727$$

$$\underline{22}$$

$$80$$

$$\underline{77}$$

$$30$$

$$\underline{22}$$

$$80$$

$$\underline{77}$$

$$3$$

$$\therefore \frac{3}{11} = 0.2727\dots = 0.\dot{2}\dot{7}$$

$$37) 95(2.56756$$

$$\underline{74}$$

$$210$$

$$\underline{185}$$

$$250$$

$$\underline{222}$$

$$280$$

$$\underline{259}$$

$$210$$

$$\underline{185}$$

$$250$$

$$\underline{222}$$

$$28$$

$$\therefore \frac{95}{37} = 2.56756\dots = 2.\dot{5}6\dot{7}$$

Desired decimal fractions are $0.\dot{2}\dot{7}$ and $2.\dot{5}6\dot{7}$

Converting repeating decimal fractions into common fractions

Example 4. Convert the following repeating decimal fractions $0.\dot{3}$, $0.\dot{2}\dot{4}$, and $42.34\dot{7}\dot{8}$ into common fractions.

Solution: In the following repeating fractions $0.\dot{3}$, $0.\dot{2}\dot{4}$ and $42.34\dot{7}\dot{8}$ have been converted into common fractions.

$$\text{First} \quad 0.\dot{3} = 0.3333 \dots$$

$$0.\dot{3} \times 10 = 0.333 \dots \times 10 = 3.333 \dots$$

$$0.\dot{3} \times 1 = 0.333 \dots \times 1 = 0.333 \dots$$

$$\text{subtracting, } 0.\dot{3} \times 10 - 0.\dot{3} \times 1 = 3$$

$$0.\dot{3} \times (10 - 1) = 3$$

$$0.\dot{3} \times 9 = 3$$

$$\therefore 0.\dot{3} = \frac{3}{9} = \frac{1}{3}$$

$$\text{Now} \quad 0.\dot{2}\dot{4} = 0.24242424 \dots$$

$$0.\dot{2}\dot{4} \times 100 = 0.242424 \dots \times 100 = 24.24242424 \dots$$

$$0.\dot{2}\dot{4} \times 1 = 0.242424 \dots \times 1 = 0.24242424 \dots$$

$$\text{subtracting, } 0.\dot{2}\dot{4} \times 99 = 24$$

$$\therefore 0.\dot{2}\dot{4} = \frac{24}{99} = \frac{8}{33}$$

$$\text{Finally} \quad 42.34\dot{7}\dot{8} = 42.34787878 \dots$$

$$42.34\dot{7}\dot{8} \times 10000 = 42.34787878 \dots \times 10000 = 423478.78787878 \dots$$

$$42.34\dot{7}\dot{8} \times 100 = 4234.7878 \dots \times 100 = 423478.7878 \dots$$

$$\text{subtracting, } 42.34\dot{7}\dot{8} \times 9900 = 423478 - 4234 = 419244$$

$$\therefore 42.34\dot{7}\dot{8} = \frac{419244}{9900} = \frac{34937}{825} = 42\frac{287}{825}$$

$$\therefore \text{desired common fractions are respectively } 0.\dot{3} = \frac{1}{3}, \quad 0.\dot{2}\dot{4} = \frac{8}{33}, \quad 42.34\dot{7}\dot{8} = 42\frac{287}{825}$$

Explanation: From the three examples above it can be seen that

- Repeating fraction is multiplied with a number equalling 1 followed by as many 0s as there are digits after decimal point of a repeating fraction.

- The repeating decimal fraction has been multiplied with a number equalling 1 followed by as many 0s as many nonrepeating digits are there in the repeating decimal fraction.
- Subtracting the second number from the first results in a whole number. It may be noted that nonrepeating part has been subtracted from the repeating decimal fraction by deleting decimal point and repeating point.
- The result of subtraction has been divided by a number equalling as many 9s as many repeating digits were there in the original repeating decimal fraction, followed by as many 0s as there were nonrepeating digits after decimal point.
- In converting repeating decimal fraction into common fraction we have denominator having as many 9s as many repeating digits followed by as many 0s as many nonrepeating digits after decimal point. Numerator equals a whole number we get after removing decimal and repeating points in the fraction less the number obtained by deleting the decimal point and all digits in the repeating part.

Remarks: Repeating decimal fractions can always be converted to common fractions. All repeating decimal fractions are rational numbers.

Example 5. Convert $5.23\dot{4}5\dot{7}$ into a common fraction.

Solution:

$$\begin{array}{rcl}
 5.23\dot{4}5\dot{7} & = & 5.23457457457\dots \\
 5.23\dot{4}5\dot{7} \times 100000 & = & 523457.457457\dots \\
 5.23\dot{4}5\dot{7} \times 100 & = & 523.457457\dots \\
 \hline
 \text{subtracting, } 5.23\dot{4}5\dot{7} \times 99900 & = & 522934 \\
 \therefore 5.23\dot{4}5\dot{7} & = & \frac{522934}{99900} = \frac{261467}{49950} = 5\frac{11717}{49950}
 \end{array}$$

$\therefore$ The sought fraction is $5\frac{11717}{49950}$.

Explanation: Since there are 5 digits in the decimal part, the repeating decimal fraction has been multiplied by 100000 (5 0s after 1). Since there are two decimal digits before the repeating part 100 (1 followed by 2 0s) repeating decimal fraction has been multiplied by 100. The second product has been deducted from the first. On one side of subtraction is a whole number, and in the other the repeating decimal number multiplied by $(100000 - 1000) = 99900$. Both sides are divided

by 99900 to obtain the common fraction.

Rules for converting repeating decimal fractions into common fractions:

Numerator of the desired fraction equals the integer obtained by deleting the decimal point and subtracting nonrepeating portion after the decimal point.

Denominator is obtained by putting as many 9s as many digits in the repeating part followed by as many 0s in the nonrepeating aprt after the decimal point.

These rules have been applied in the following examples to convert a repeating decimal fractions into common fractions.

Example 6. Convert $45.\dot{2}\dot{3}4\dot{6}$ into common fraction.

$$\text{Solution: } 45.\dot{2}\dot{3}4\dot{6} = \frac{452346 - 452}{9990} = \frac{451894}{9990} = \frac{225947}{4995} = 45\frac{1172}{4995}$$

$\therefore$ the desired fraction is $45\frac{1172}{4995}$.

Example 7. Convert $32.\dot{5}\dot{6}\dot{7}$ into common fraction.

$$\text{Solution: } 32.\dot{5}\dot{6}\dot{7} = \frac{32567 - 32}{999} = \frac{32535}{999} = \frac{3615}{111} = \frac{1205}{37} = 32\frac{21}{37}$$

$\therefore$ the desired fraction is $32\frac{21}{37}$.

Work: Convert $0.\dot{4}\dot{1}$, $3.04\dot{6}2\dot{3}$, $0.0\dot{1}\dot{2}$ and $3.31\dot{2}\dot{4}$ into common fractions.

Similar and dissimilar repeating decimal fractions

If for two or more repeating decimal fractions number of digits both in repeating and nonrepeating parts are the same then they are called similar repeating decimal fractions. Otherwise they are called dissimilar repeating decimal fractions. For example, $12.\dot{4}\dot{5}$ and $6.\dot{3}\dot{2}$; $9.45\dot{3}$ and $125.89\dot{7}$ are similar repeating decimal fractions. But, $0.34\dot{5}\dot{6}$ and $7.45\dot{7}8\dot{9}$; $6.43\dot{5}\dot{7}$ and $2.89\dot{3}4\dot{5}$ are dissimilar repeating decimal fractions.

Converting dissimilar repeating decimal fractions into similar ones

If we write repeating part of a repeating decimal fraction the value does not change. For example $6.45\dot{3}\dot{7} = 6.45\dot{3}7\dot{3}\dot{7} = 6.453\dot{7}\dot{3} = 6.4537\dot{3}\dot{7}$. Here each one of them is a repeating decimal fraction $6.45373737\dots$, which is an infinite decimal

number. If we convert this repeating decimal number into common fraction we will see that all of them are equal.

$$\begin{aligned} 6.45\dot{3}\dot{7} &= \frac{64537 - 645}{9900} = \frac{63892}{9900} \\ 6.45\dot{3}73\dot{7} &= \frac{6453737 - 645}{999900} = \frac{6453092}{999900} = \frac{63892}{9900} \\ 6.45373\dot{7} &= \frac{6453737 - 64537}{990000} = \frac{6389200}{990000} = \frac{63892}{9900} \end{aligned}$$

In order to convert into similar repeating decimal fractions every repeating decimal must have equal number of digits in nonrepeating part. Repeating part must have number of digits equal to lowest common multiple of digits in all repeating parts.

Example 8. Convert $5.\dot{6}$, $7.3\dot{4}\dot{5}$, and $10.78\dot{4}2\dot{3}$ into similar repeating decimal fractions.

Solution: Repeating decimal fractions $5.\dot{6}$, $7.3\dot{4}\dot{5}$, and $10.78\dot{4}2\dot{3}$ have nonrepeating digits equalling 0, 1 and 2 respectively. Here $10.78\dot{4}2\dot{3}$ has the most number of digits in nonrepeating part, so in order to convert into similar repeating decimal fractions we must have 2 digits in nonrepeating part of every number. $5.\dot{6}$, $7.3\dot{4}\dot{5}$ and $10.78\dot{4}2\dot{3}$ repeating decimal fractions have respectively 1, 2 and 3 digits in the repeating part. LCM of 1, 2 and 3 is 6. So in order to convert into similar repeating decimal fractions we must have 6 digits in the repeating part of every number. So, $5.\dot{6} = 5.6\dot{6}\dot{6}\dot{6}\dot{6}\dot{6}\dot{6}$, $7.3\dot{4}\dot{5} = 7.3\dot{4}\dot{5}\dot{4}\dot{5}\dot{4}\dot{5}\dot{4}$ and $10.78\dot{4}2\dot{3} = 10.78\dot{4}23\dot{4}2\dot{3}$. Desired similar repeating decimal fractions are respectively $5.6\dot{6}\dot{6}\dot{6}\dot{6}\dot{6}\dot{6}$, $7.3\dot{4}\dot{5}\dot{4}\dot{5}\dot{4}\dot{5}\dot{4}$ and $10.78\dot{4}23\dot{4}2\dot{3}$.

Example 9. Convert 1.7643 , $3.\dot{2}\dot{4}$ and $2.78\dot{3}4\dot{6}$ similar repeating decimal fractions.

Solution: In 1.7643 we have 4 digits in the nonrepeating part and it does not have repeating part. In $3.\dot{2}\dot{4}$ number of digits in nonrepeating part is 0 and that of repeating part is 2; in $2.78\dot{3}4\dot{6}$ number of digits in nonrepeating part is 2 and that in repeating part is 3. Here the largest number of nonrepeating digits is 4 and number of digits in the repeating parts are 2 and 3, LCM of which is 6. So each number must have 4 nonrepeating digits and 6 repeating digits.

$$\therefore 1.7643 = 1.7643000000, 3.\dot{2}\dot{4} = 3.2424242424 \quad 2.78\dot{3}4\dot{6} = 2.7834\dot{6}34\dot{6}3\dot{4}$$

Desired similar repeating decimal fractions are respectively 1.7643000000 , 3.2424242424 and $2.7834\dot{6}34\dot{6}3\dot{4}$

Remarks: In order to convert finite decimal fractions into similar decimal fractions we must add sufficient number of 0s at the end of the number to match with number of nonrepeating digits. In case of repeating decimal fractions for each number, numbers of nonrepeating digits have been made equal so should be the case with repeating digits.

Work: Convert 3.467 , $2.01\dot{2}4\dot{3}$ and $7.52\dot{5}\dot{6}$ into similar repeating decimal fractions.

Addition and subtraction of repeating decimal fractions

In order to add repeating decimal fractions we need to convert them into similar repeating decimal fractions. Then addition and subtraction must be carried out as in case of finite decimal fractions. Results thus found will not necessarily be the right results. In order to obtain the correct value we must add or subtract from the last digit of the result, the digit in hand when the leftmost repeating digits were added/subtracted.

Remark:

1. Addition and subtraction of repeating decimal fractions result in repeating decimal fractions. In addition and subtraction result is yet another repeating decimal fraction having number of nonrepeating digits equalling the largest/larger number of nonrepeating digits among the numbers involved in the operation. Number of digits in the repeating part equals LCM of the numbers of digits in the repeating part. In case of finite decimal fractions the number of digits in the nonrepeating part is matched with the largest such number among numbers involved in addition or subtraction by appending appropriate number of 0s at the end.
2. We can also convert repeating decimal fractions into common fractions and then do addition or subtraction. In this method more addition or subtraction operations will be necessary.

Example 10. Add $3.\dot{8}9$, $2.1\dot{7}\dot{8}$ and $5.8979\dot{8}$.

Solution: We have 2 digits in nonrepeating part, and repeating parts have respectively 2, 2 and 3 for which LCM is 6. At first all these three repeating decimal fractions have been converted into similar repeating decimal fractions.

$$\begin{array}{r}
3.\dot{8}\dot{9} \quad = 3.89\dot{8}9898\dot{9} \\
2.1\dot{7}\dot{8} \quad = 2.17\dot{8}7878\dot{7} \\
5.89\dot{7}9\dot{8} \quad = 5.89\dot{7}9879\dot{8} \\
\hline
11.97576574 \quad [8 + 8 + 7 + 2 = 25, \text{ here 2 is 2 in hands} \\
\quad \quad \quad +2 \quad \text{here 2 of 25 has been added}] \\
\hline
11.97\dot{5}7657\dot{6}
\end{array}$$

Desired result is $11.97\dot{5}7657\dot{6}$ or, $11.97\dot{5}7\dot{6}$

Remarks: In the result 576576 is repeating part. But if we assume 576 is the repeating part then there will be no change of value .

Note: In order to clarify addition at the rightmost digit we have solved the addition in a different way:

$$\begin{array}{r}
3.\dot{8}\dot{9} \quad = 3.89\dot{8}9898\dot{9}|89 \\
2.1\dot{7}\dot{8} \quad = 2.17\dot{8}7878\dot{7}|87 \\
5.89\dot{7}9\dot{8} \quad = 5.89\dot{7}9879\dot{8}|79 \\
\hline
11.97\dot{5}7657\dot{6}|55
\end{array}$$

Here after the end of repeating part some more digits have been taken. These extra digits have been separated by a vertical line. The addition has been performed. From the addition of digits just after the vertical line 2 in hands has been added to the digits just before the vertical line. It may be noted here the digit on the right side of the vertical line is the same as the first digit of repetition.

Example 11. Add $8.94\dot{7}\dot{8}$, 2.346 $4.\dot{7}\dot{1}$.

Solution: In order to make repeating decimal fractions similar we must have 3 nonrepeating digits and repeating part must have digits equal to LCM of 2 and 3, that is 6. Now decimal fractions will be added.

$$\begin{array}{r}
8.94\dot{7}\dot{8} \quad = 8.947\dot{8}4784\dot{7} \\
2.346 \quad = 2.346000000 \\
4.\dot{7}\dot{1} \quad = 4.717\dot{1}717\dot{1} \\
\hline
16.011019564 \quad [8 + 0 + 1 + 1 = 10, \text{ Here 1 is 1 in hand} \\
\quad \quad \quad +1 \quad \text{Here 1 of 10 has been added}] \\
\hline
16.011\dot{0}1956\dot{5}
\end{array}$$

The desired sum is $16.011\dot{0}1956\dot{5}$

Work: Add 1) $2.09\dot{7}$ $5.12\dot{7}6\dot{8}$ 2) $1.34\dot{5}$, $0.31\dot{5}7\dot{6}$ $8.056\dot{7}\dot{8}$

Example 12. Subtract $5.24\dot{6}7\dot{3}$ from $8.24\dot{3}$.

Solution: Here number of digits in nonrepeating part should be 2 and that in repeating part should be 6, LCM of 2 and 3. Now the numbers have been converted into similar numbers, and subtraction has been performed as follows:

$$\begin{array}{r}
 8.2\dot{4}\dot{3} = 8.2\dot{4}\dot{3}434\dot{3}\dot{4} \\
 5.2\dot{4}\dot{6}\dot{7}\dot{3} = 5.2\dot{4}\dot{6}\dot{7}367\dot{3} \\
 \hline
 2.99\dot{6}6976\dot{1} \quad [\text{subtracting 6 from 3 results in 1 in hand}] \\
 \quad \quad \quad -1 \\
 \hline
 2.99\dot{6}6976\dot{0}
 \end{array}$$

The desired result is 2.99669760.

Remarks: If the minuend starting at the repeating dot is smaller than the subtrahend we must subtract 1 from the rightmost digit.

Note: In the following is explained why 1 is subtracted from the rightmost digit.

$$\begin{array}{r}
 8.2\dot{4}\dot{3} = 8.2\dot{4}\dot{3}434\dot{3}\dot{4}|34 \\
 5.2\dot{4}\dot{6}\dot{7}\dot{3} = 5.2\dot{4}\dot{6}\dot{7}367\dot{3}|67 \\
 \hline
 2.99\dot{6}6976\dot{0}|67
 \end{array}$$

The desired result is 2.99669760|67. We have obtained the same results in both the cases.

Example 13. Subtract $16.\dot{4}3\dot{7}$ from $24.45\dot{6}4\dot{5}$.

Solution:

$$\begin{array}{r}
 24.45\dot{6}4\dot{5} = 24.45\dot{6}4\dot{5} \\
 16.\dot{4}3\dot{7} = 16.43\dot{7}4\dot{3} \\
 \hline
 8.01902 \quad [\text{if 7 is subtracted from 6 we have 1 in hand}] \\
 \quad \quad \quad -1 \\
 \hline
 8.0190\dot{1}
 \end{array}$$

The desired result is 8.01901.

Note: In the following is explained why 1 is subtracted from the rightmost digit.

$$\begin{array}{r}
 24.45\dot{6}4\dot{5} = 24.45\dot{6}4\dot{5}|64 \\
 16.\dot{4}3\dot{7} = 16.43\dot{7}4\dot{3}|74 \\
 \hline
 8.0190\dot{1}|90
 \end{array}$$

Work: Subtract 1) Subtract 10.418 from $13.12\dot{7}8\dot{4}$
 2) Subtract $9.12\dot{6}4\dot{5}$ from $23.039\dot{4}$

Multiplication and division of repeating decimal fractions

If repeating decimal fractions are first converted into common fractions and then multiplication or division operations are performed we can convert the result into repeating decimal fractions. If finite decimal fractions and repeating decimal fractions are multiplied or divided then this rule should be followed. However, in case of division if both divisor and dividend are repeating decimal fractions then converting them into similar repeating decimal fractions will ease the operations.

Example 14. Multiply $4.\dot{3}$ with $5.\dot{7}$.

Solution:

$$\begin{aligned} 4.\dot{3} &= \frac{43 - 4}{9} = \frac{39}{9} = \frac{13}{3} \\ 5.\dot{7} &= \frac{57 - 5}{9} = \frac{52}{9} \\ \therefore 4.\dot{3} \times 5.\dot{7} &= \frac{13}{3} \times \frac{52}{9} = \frac{676}{27} = 25.\dot{0}3\dot{7} \end{aligned}$$

The desired result is $25.\dot{0}3\dot{7}$.

Example 15. Multiply $0.2\dot{8}$ with $42.\dot{1}8$.

Solution:

$$\begin{aligned} 0.2\dot{8} &= \frac{28 - 2}{90} = \frac{26}{90} = \frac{13}{45} \\ 42.\dot{1}8 &= \frac{4218 - 42}{99} = \frac{4176}{99} = \frac{464}{11} \\ \therefore 0.2\dot{8} \times 42.\dot{1}8 &= \frac{13}{45} \times \frac{464}{11} = \frac{6032}{495} = 12.1\dot{8}\dot{5} \end{aligned}$$

The desired result is $12.1\dot{8}\dot{5}$.

Example 16. How much is $2.5 \times 4.3\dot{5} \times 1.2\dot{3}4\dot{?}$